Week 20 — Extended Reading: Group advantages and verifiable rewards

STUDY-RL

Source audit: 20 October 2026

Read for a claim, an assumption and a test

Read the teaching guide first, then use this pack alongside the lecture. The annotations below are course-written paraphrases and investigation prompts, not quotations from the papers. Each source link identifies a primary paper or official guidance. Benchmark results do not transfer automatically to our small exercises. For each reading, write the mathematical object, the needed assumptions and one observation that would challenge your interpretation.

1 Shao et al. (2024): DeepSeekMath / GRPO

Primary source. https://arxiv.org/abs/2402.03300

Mechanism and scope. GRPO uses groups of responses to estimate relative advantages without a separate value critic. Group mean and standard-deviation normalisation affect both credit and sample weighting. The policy ratio and reference KL have different roles.

Read, derive, investigate. For rewards $(1, 0, 1, 0)$, calculate the mean and population standard deviation. What happens when every reward in the group is equal? State the epsilon convention and distinguish coefficient variance from the variance of the complete policy-gradient estimator.

2 Ahmadian et al. (2024): RLOO

Primary source. https://arxiv.org/abs/2402.14740

Mechanism and scope. RLOO revisits REINFORCE-style optimisation with the other sampled responses as a baseline. Conditional independence of those samples lets the leave-one-out baseline avoid direct dependence on the current sampled action.

Read, derive, investigate. Show $A_i^{\mathrm{LOO}} = G(r_i - \bar{r})/(G-1)$ for $G > 1$. For the four rewards above, obtain $\pm 2/3$. Their sum is exactly zero; larger coefficient scale does not by itself prove worse gradient variance or inferior learning.

Connect to the proof. Open the lecture's RLOO unbiasedness theorem. Identify the step where conditional independence makes the leave-one-out baseline term vanish in expectation. Then include the current response in the baseline and derive the $(G-1)/G$ scaling of the expected on-policy gradient. State why neither calculation proves a universal variance reduction.

3 Liu et al. (2025): Understanding R1-Zero-like training

Primary source. https://arxiv.org/abs/2503.20783

Mechanism and scope. The Dr.GRPO analysis identifies effects of response-length and question-difficulty weighting in common GRPO normalisations. Its ablations motivate inspecting the exact reduction order in an implementation.

Read, derive, investigate. Write per-token and per-sequence sums before averaging. Which examples receive more total weight? Change only one normalisation and compare a fixed batch gradient; distinguish that controlled calculation from a full training claim.

4 DeepSeek-AI (2025): DeepSeek-R1

Primary source. https://arxiv.org/abs/2501.12948

Mechanism and scope. The report separates R1-Zero from a multi-stage R1 recipe and studies reinforcement learning for reasoning. In the cited R1-Zero training description, the KL coefficient is nonzero; it is not evidence that all successful reasoning RL removes reference regularisation.

Read, derive, investigate. Identify the training stage, initial checkpoint, reward and evaluation for every claimed result. Longer outputs or an appealing example do not alone establish general reasoning ability. The small integer task in this repository has a much narrower verifier and action space.

5 Lightman et al. (2023): Process supervision

Primary source. https://arxiv.org/abs/2305.20050

Mechanism and scope. The paper compares process and outcome supervision and introduces PRM800K human step-quality labels. A learned process reward is an estimate of step quality, not a proof checker.

Read, derive, investigate. Distinguish human step labels from automatically generated continuation labels. Construct a step that looks plausible locally but creates a later contradiction. Explain how your evaluation would detect it without consulting the same model that supplied the reward.

Connect the reading to your workbook

Start with exact arithmetic and malformed-answer tests before running optimisation. Keep prompt parsing separate from completion parsing. A deterministic verifier is reproducible, but its accepted language defines exploitable edges. Report valid-answer accuracy, parser failures, response length and reference divergence separately; a reward total alone cannot reveal which mechanism changed.
Evidence boundary. Source identities and the central claims above were checked against primary material. These annotations do not reproduce every proof, training run or empirical result. Use the original paper for theorem assumptions and experimental details; use `project/` and the workbook for the exact course implementation.

Statistics reading bridge

Zero-sum coefficients do not imply a zero policy gradient. Read variance of products/sums with the complete estimator visible. Reconstruct one RLOO baseline identity, then compare full score-weighted gradient samples over independent groups; coefficient variance alone cannot rank full estimators.

Checked textbook locations: Siegrist § § 4.3, 4.5, 6.2, PDF pp.460, 478, 817. https://stats.libretexts.org/@go/page/10160. These locations support the probability and estimation concepts; the algorithm-specific results remain attributed to the primary readings above. The course source catalog distinguishes the original textbook from its supplied instructor manual and records PDF-viewer pagination.